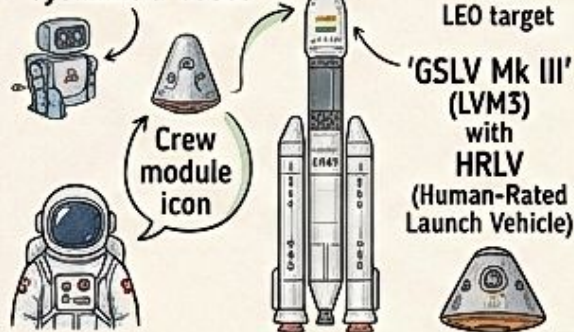


ISRO'S MAJOR 2026 MISSIONS ROADMAP

★ GAGANYAAN

(2026, Human Spaceflight)

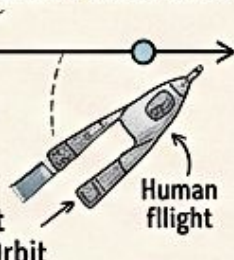
- 'Vyommitra' robot



GAGANYAAN-2/3 (Uncrewed),
GAGANYAAN-1 (Crewed - late 2026)

- Milestones for orbital
- Docking, & Finally
- Human flight to Low Earth Orbit

- Milestones for orbital
- Docking
- A human flight to Low Earth Orbit



★ NISAR

(NASA-ISRO Synthetic Aperture Radar)

- 'L & S Band Radar'
- Earth Observation
- Mapping ice, forests

★ SPADEX

(Space Docking Experiment)

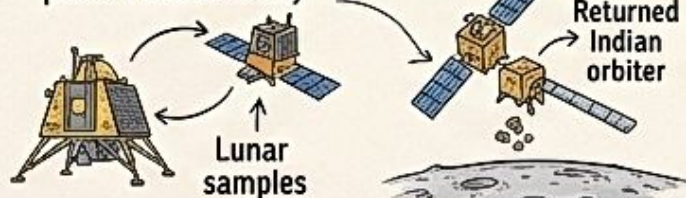
- 'Docking Technology'
- 'In-orbit Assembly'



- Precursor to Space Station

★ CHANDRAYAAN-4

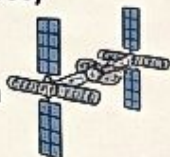
(Sample Return, planned late 2027/28)



★ BHARATIYA ANTARIKSH STATION (BAS)

(By 2035)

- Planned Indian Space Station



★ ADITYA-L1

(Current, for context)

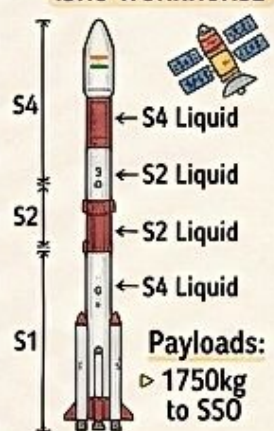
- Solar L1
- Studying Sun
- Sun



★ LAUNCH VEHICLE ARCHITECTURE (PSLV vs GSLV/LVM3)

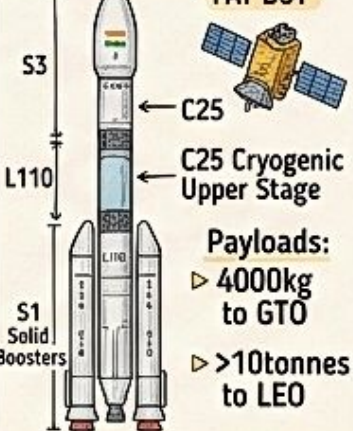
PSLV
(Polar Satellite Launch Vehicle)

"ISRO WORKHORSE"



GSLV/LVM3
(Geosynchronous Launch Vehicle / Launch Vehicle Mark-3)

"FAT BOY"



★ CRYOGENIC ENGINE TECHNOLOGY (C25/CE-20)



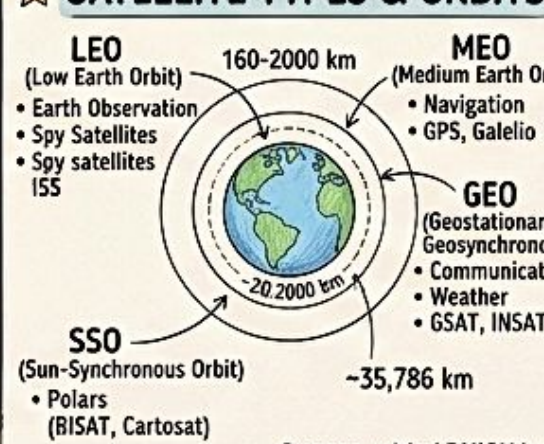
★ NAVIGATION SYSTEMS

(NavIC / IRNSS)



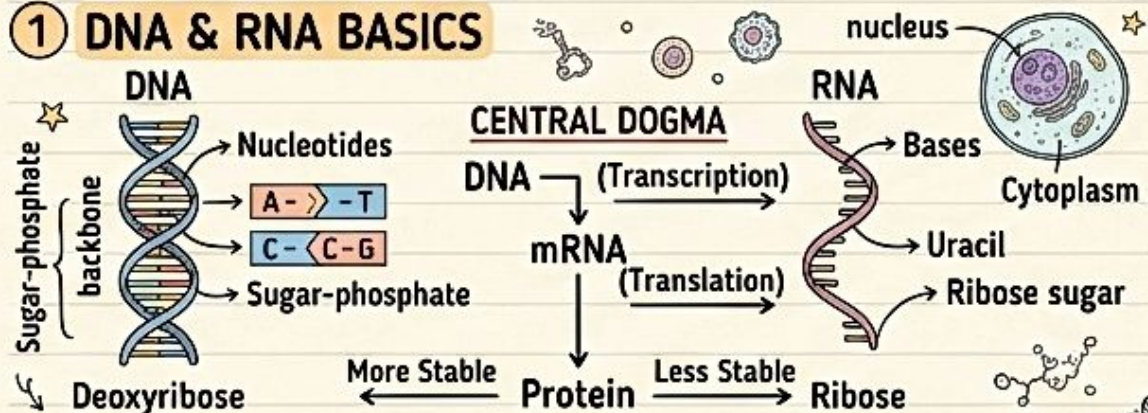
- 'REGIONAL' (India + 1500km)
- Independent GPS

★ SATELLITE TYPES & ORBITS

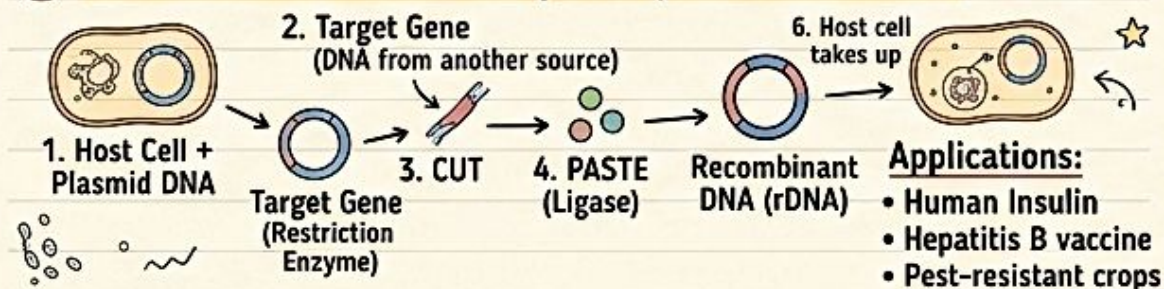


BIOTECHNOLOGY FOR UPSC PRELIMS 2026

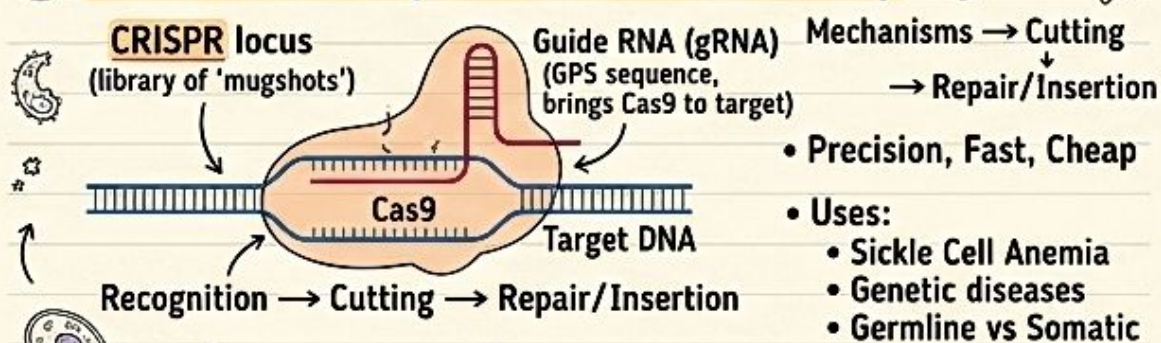
1 DNA & RNA BASICS



2 RECOMBINANT DNA (rDNA) TECHNOLOGY

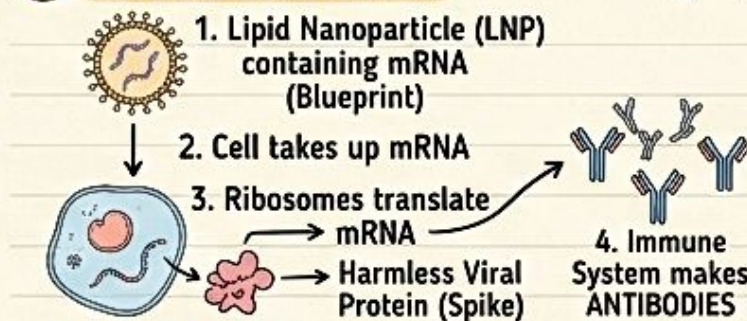


3 CRISPR-Cas9 (SEARCH & REPLACE)



ABHISHA AI
by Abhishek Pathak

4 mRNA VACCINES



Advantages:

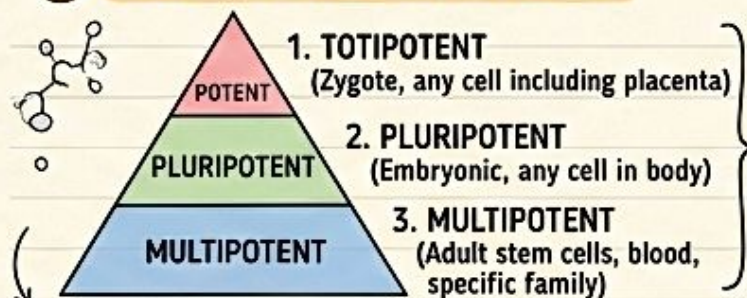
- Rapid development
- No live virus

Challenge:

- Temperature stability (cold chain)

E.g., Pfizer-BioNTech, Moderna for COVID-19

5 STEM CELLS & POTENCY



iPSCs

(Induced Pluripotent Stem Cells)

- Re-programmed somatic cells to act like ESCs (avoids ethical issues)

Uses: Drug testing, disease modeling, regenerative medicine

6 GM CROPS & BIOFORTIFICATION

GM Crops	Biofortification
Method: Transgenic, foreign gene Goal: Pest/Herbicide Tolerance Examples: • Bt Cotton (Pest) • Bt Brinjal • DMH-11 Mustard (Herbicide Tolerance)	Method: Breeding/GE Goal: Nutritional improvement Examples: • GOLDEN RICE (Vitamin A) • Iron-rich Millet

GEAC (under MoEF&CC, apex body)
Current India status: Bt Cotton only commercial

S&T ESSENTIALS: DEEP TECH FOR UPSC PRELIMS 2026

ARTIFICIAL INTELLIGENCE (AI) & MACHINE LEARNING (ML)



- **CONCEPT:** Simulate human intelligence. ML: Systems that learn from data. DL: Neural Networks (like human brain).

- ★ **KEY TERMS:** NLP (Speech, Text), Computer Vision (Images), Turing Test.

GENERATIVE AI: Large Multimodal Models (LMMs), 'Agentic AI' (Goal-driven).



- ★ **INDIA'S PILLARS:** IndiaAI Mission (Compute, Datasets), AIRAWAT Supercomputer.

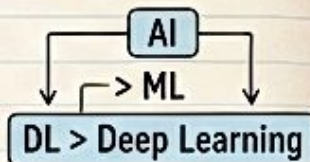
CYBERSECURITY



- ★ **THREAT LANDSCAPE:** Ransomware-as-a-Service (RaaS), Deepfakes (AI misinformation), Zero-Day Vulnerabilities.
- ★ **INDIA'S DEFENSE:** CERT-In (Emergency Response), National Cyber Security Policy.
- ★ **CONCEPTS:** Zero Trust Architecture (Never trust, always verify), Multi-factor Authentication (MFA).



HIERARCHY



QUANTUM COMPUTING

QUBITS

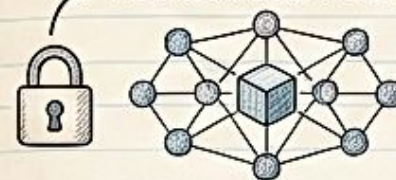
- ★ **BASIC UNIT:** Qubits (non-classical).
- ★ **PRINCIPLES:** Superposition, Entanglement (instant influence).
- ★ **'Q-DAY' RISK:** Threat to RSA Encryption → Post-Quantum Cryptography (PQC).
- ★ **NATIONAL QUANTUM MISSION (NQM):** Developing 50-1000 qubits by 2031.



BLOCKCHAIN TECHNOLOGY



DECENTRALIZED LEDGER

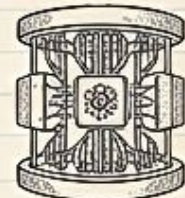


- ★ **MECHANISM:** Proof of Work vs. Proof of Stake, Hashing, Cryptography.
- ★ **APPLICATIONS:** Smart Contracts (self-executing code), NFTs (unique assets), CBDC (RBI's e-Rupee).

- ★ **WEB 3.0:** User-owned decentralized internet.

SUPERPOSITION

ENTANGLEMENT



5G & COMMUNICATION TECH

- ★ **5G FEATURES:** High Speed, Ultra-low Latency, Massive Machine-Type Communications (mMTC).
- ★ **NETWORK SLICING:** Virtual networks on one infrastructure.
- ★ **SMALL CELLS:** Essential for coverage.
- ★ **BEYOND 5G:** Bharat 6G Vision, Edge Computing (Process data near source).

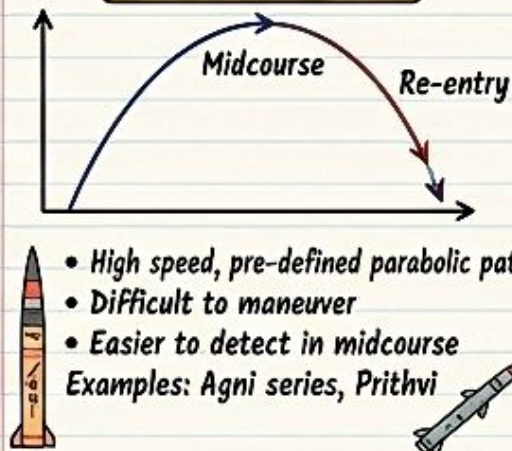


NUCLEAR FISSION VS. FUSION

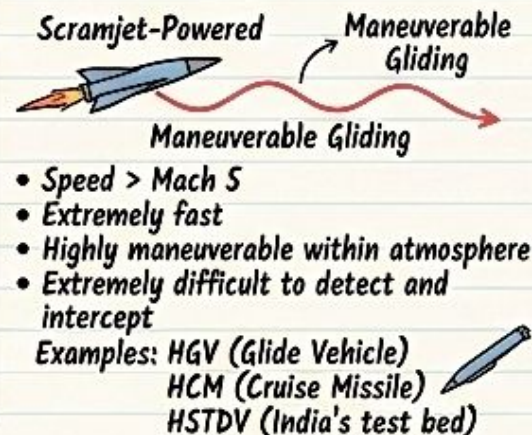
FISSION	FUSION
 U-235	 D, T
- Splitting heavy nucleus into lighter ones (Uranium, Plutonium) - releases significant energy - requires chain reaction control - used in current power plants & Atomic Bombs - produces radioactive waste	- Combining light nuclei into heavier ones (Hydrogen isotopes) - releases immense energy - requires extreme heat & pressure - fuels stars/Sun - still experimental (e.g., ITER) - minimal radioactive waste - uses readily available fuel

BALLISTIC VS. HYPERSONIC TECHNOLOGY

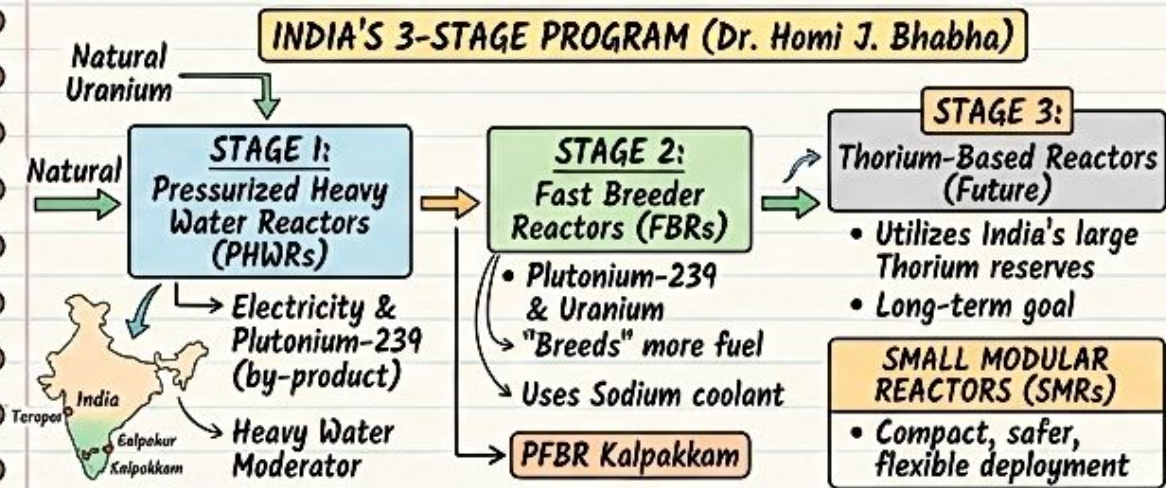
BALLISTIC MISSILE



HYPERSONIC MISSILE



NUCLEAR REACTORS & INDIA'S PROGRAM



DEFENCE RESEARCH ORGANIZATIONS

DRDO (Defence Research & Development Organisation)

- Nodal agency for indigenous defense technology
- Missiles, LCA Tejas, electronic warfare
- Recent: MISSION DIVYASTRA (Agni-V with MIRV)

HAL (Hindustan Aeronautics Limited)

- Focuses on military aircraft & helicopters
- LCA (Light Combat Aircraft)
- LCH (Light Combat Helicopter)

DAE (Department of Atomic Energy)

- Oversees nuclear energy & research (under PMO)
- BARC (Bhabha Atomic Research Centre)

HAL (Hindustan Aeronautics Limited)

- Focuses on military tank & helicopters
- LCA (Light Combat Aircraft)

IDEX (Innovations for Defence Excellence)

- Ecosystem for startups & MSMEs in defense

UPSC PRELIMS 2026: SCIENCE & TECHNOLOGY QUICK REFERENCE GUIDE

VACCINE TECHNOLOGIES

MRNA VACCINES



- synthetic mRNA
- spike protein instruction
- Pfizer/Moderna

VIRAL VECTOR



- modified virus
- genetic code delivery
- Covishield

INACTIVATED

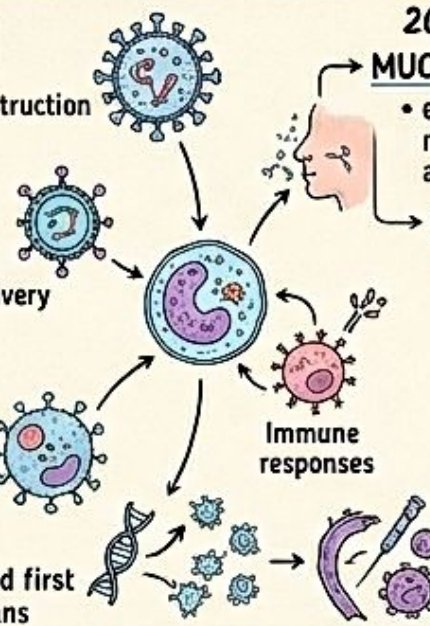


- killed germ
- Covaxin

DNA VACCINES



- ZyCoV-D - India's world first plasmid DNA for humans



2026 FOCUS -

MUCOSAL (NASAL) VACCINES

- e.g., iNOVACC, administered in nasal passages to induce systemic and mucosal immune responses

COMBINATION VACCINES

- Combination vaccines contain multiple antigens in a single dose, e.g., MR, MMR, DPT, and hexavalent vaccines, to provide protection against several diseases.

High-Yield Fact:

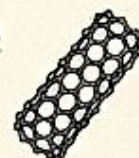
Prelims focus: Mucosal vaccines induce localized (mucosal) immunity, crucial for respiratory pathogens.

NANOTECHNOLOGY

KEY MATERIALS

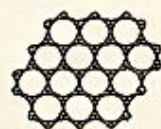
CARBON NANOTUBES

- High strength
- Conductive
- Lightweight armor



GRAPHENE

- Single layer carbon
- Wonder material
- Flexible
- Wonder material- flexible
- Monot and healthcorpies



SECTOR APPLICATIONS

NANO-MEDICINE

- Targeted Drug Delivery
- Nanobots in surgery → in surgery



NANO-AGRICULTURE

- Nano-fertilizers (e.g. IFFCO Nano Urea)



QUANTUM DOTS

- Displays, imaging
- Quanting containents



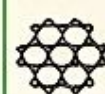
KEY MATERIALS

CARBON NANOTUBES



- high strength
- Conductive
- Lightweight armor

GRAPHENE



- single layer carbon
- wonder material
- Flexible

ROBOTICS

TYPES

AMRs

- Autonomous Mobile Robots

HUMANOIDS

- Vyommitra for ISRO Gaganyaan

EXOSKELETONS

- military, rehab



KEY COMPONENTS

ACTUATORS ('muscles')



SENSORS ('senses')



MICROCONTROLLERS ('brain')



SWARM ROBOTICS

- cooperative behavior
- disaster management



RENEWABLE ENERGY TECHNOLOGIES

CURRENT STATUS

- 2026 Milestone: 50% non-fossil installed capacity



KEY TECHS

SOLAR PV

- direct conversion
- concentrated heat

vs.

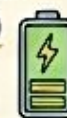
SOLAR THERMAL

- direct conversion vs.
- concentrated heat



OFFSHORE WIND

- Gujarat, Tamil Nadu focus



ENERGY STORAGE

- Solid-state batteries
- Pumped Hydro

HYDROGEN FUEL

THE COLOR SPECTRUM

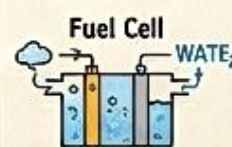
GREEN	Renewable + Electrolysis = Zero Emissions
BLUE	Natural Gas + Carbon Capture
GREY	Natural Gas, High CO ₂
BROWN	Coal, highly polluting

NATIONAL GREEN HYDROGEN MISSION

- At least 5 MMT production per annum by 2030

SIGHT PROGRAM

- Incentives for electrolyzers/production



PREPARED WITH
ABHISHA AI
by Abhishek Pathak